

Planetary Chemistry: The Quest for Iron Oxide

Teaching with AI (TAI)

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Subject: Chemistry

Activity duration: 120 min

Phases:

1. Choice and characterization of exoplanet sample with AI-lab (30 min)
2. Colorimetry in the real lab to get an approximate concentration (50 min)
3. Filtering and weighing the iron oxide (40 min)

Learning goals:

- Learn about the characterization of substances with an AI lab that can perform fast, small experiments embedded in the exoplanet context.
- Analytical chemistry: learn how to do a colorimetry to estimate a concentration.
- Learn techniques such as filtering or dehydration
- Work with stoichiometric relations of chemistry reactions and compute masses and solution concentrations

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A chemistry lesson with AI

Challenge: The Aurora Network is a system of scientific stations spread across several exoplanets that monitor climate, geology, and possible signs of life. After a powerful radiation storm damaged one station, engineers need to manufacture new **iron-oxide-rich** ceramic bricks to restore the protective shielding around sensitive instruments. Because transporting construction materials from Earth is slow and costly, scientists are searching for a local source of iron at the exoplanets. Your mission is to use a rover and its AI laboratory to identify the planet most likely to contain iron-rich compounds and determine which sample could produce the greatest amount of iron oxide (Fe_2O_3). The results of your laboratory analysis will help decide whether the deposit can provide enough material to repair the station's shielding.

Assignment 1: Send a sample to Earth

You can control an AI-embedded lab inside the rover that is exploring your assigned exoplanet. The rover should take a sample and send it to Earth for further analysis. Your goals in this assignment are:

- Choose which site to take a sample from.
- Identify which is the most likely substance in the sample.

Go to the link provided by the teacher that will take you directly to the AI-lab of your corresponding rover. How to use it:

- The rover has a maximum energy of 100 points. Going to a site costs 3 points. Doing an experiment costs 10 points. Once the rover is out of energy, you won't be able to do anything else.
- Once you have moved the rover to a site, you can choose experiments in the drop-down menu. The AI will then give you the result of the chosen experiment for a sample taken from that site.
- You can chat with the AI for free in the Chat log, so don't hesitate in asking as many questions as you want.

Initial ideas

Question 1: Look at the overview image with all the sites. Which ones do you think can contain an iron compound and why?

Constructing new ideas

Question 2: In the box below, write the conclusions you get after interacting with the AI lab on each of the sites:

Site	Compound	Why? Conclusions from experimental results

Solution: For each exoplanet, look into its corresponding dossier to get the answers to this question (`https://scientificailab.streamlit.app/RoverLab`, then go to the dropdown menu "Full scenario details". Never share this webpage with the students! The students should get their own link.). In the dossier, there is a detailed explanation of each site, which substance it contains, and how the substance reacts to each of the experiments. See the additional "Teacher Guide" for more details.

Question 3: Which site have you finally chosen to get a sample and send it to Earth? Which substance do you think it contains?

Solution: They should choose Site 1 (for all exoplanets). What they get there is an Iron (III) chloride solution with different concentration for each exoplanet. Some planets have here a decoy solution (it seems it is Iron (III) chloride but it is not). Students only know about this when they do the reaction with the KOH.

Assignment 2: Analyzing the sample's concentration at the Earth's lab

The sample has arrived at the Earth's lab. Here, you will have to further analyze it to later conclude which exoplanet has the largest potential to get the biggest amount of iron oxide. Your goal now is:

- Do a colorimetry to estimate the sample concentration.

Initial ideas

Question 4: Ask the AI (or look for information elsewhere on the internet) about the possible ways that one can get Fe_2O_3 from the substance of the sample you sent to Earth. Write all the steps of the process here, including the chemical reactions (properly balanced!). Note that you need to do these steps with substances that are available at your lab.

Solution: See Teachers Guide for details and for the properly balanced reactions. In summary:

1. Make $FeCl_3$ react with KOH to get $Fe(OH)_3$ (precipitated).
2. Filtrate the precipitated $Fe(OH)_3$.
3. Thermally decompose the hydroxide $Fe(OH)_3$ into an oxide Fe_2O_3

Question 5: Why do you need to estimate the sample concentration before doing anything else?

Solution: To know how much KOH to add in order to get all the Iron(III) chloride to react (in turn, to extract as much iron oxide as possible from the sample).

Question 6: How would you estimate the concentration (in moles of solute/L of solution) of the sample solution you got (considering that you cannot distil the water out)?

Solution: Getting to the idea of colorimetry. Before the next step, students should understand how the colorimetry works.

Constructing new ideas

Do the colorimetry of the sample to get its estimated concentration. If there is extra time, you can do all the steps of the colorimetry yourself. To do that, first, you need to create several control samples of which you know the concentration, to then compare the color of the exoplanet sample to the color of the control samples and estimate its concentration.

In case there is no extra time, the teacher will have the control samples prepared for you, so you would just need to take your sample and compare it to those control ones to estimate the concentration. In this case, skip the following question and answer the next one.

Question 7: Make a list of the concentrations you created for the control samples.

Question 8: Which is your best estimation of the exoplanet's sample concentration based on the colorimetry?

Assignment 3: Getting the iron (III) oxide and measuring its mass

As you have seen, doing the colorimetry is not a very precise measurement for how much iron is in the sample. But we need this initial measurement for our next step. We now are going to extract the iron oxide and measure it. Your goals now are:

- Do the chemical reactions needed to get the iron oxide Fe_2O_3.
- Measure the final mass of iron oxide you get from the sample.

Question 9: Look again at the answer of Question 4. How much KOH do you need to add to your exoplanet sample to make sure that *all* the $Fe(Cl)_3$ precipitates? Write the calculations here.

Look again at the answer of Question 4 for the steps and chemical reactions you are about to do.

- **Do the reaction with the KOH.**
- **Afterwards, filter the precipitated substance and heat it up.**
- **Weigh the resulting substance.**

Question 10: Describe the resulting substance. Is it what you expected to get in terms of color, consistency, etc. (you can check with an AI what is the expected aspect of the Fe_2O_3)? If not, what do you think has happened?

Solution: See Teachers Guide.

Question 11: Compute the mass of Fe_2O_3 you expected to get if the concentration of the sample was the one you estimated with the colorimetry. Is the one you actually got larger or smaller?

Solution: See Teachers Guide for the expected masses for each of the samples.

Closing the challenge

Write in your exoplanet file the resulting final weight of the iron oxide.

What have we learned?

Write in your file the things you have learned in this lesson (Reflections) and how you used AI for science in this lesson (AI assistance).

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